

Developing a Low-Cost Wireless Transtibial Prosthetic Socket Sensor Sock: An Iterative Design Study

A. W. Qureshi, N. Alam, H. Tariq, K. Chen, A. Wong
Project90, Schulich School of Engineering, University of Calgary

Introduction

The Prosthetic fitment process remains a major issue in patient rehabilitation, with 48% of patients and 65% of clinicians listing it as the biggest concern [1]. The current fitting method is iterative, depending on qualitative patient feedback, visual inspection, and extensive clinical experience. Few solutions that provide quantitative data on limb-socket interface pressure exist as a result of cost and comfort. Creating a practical solution would provide an objective standard of socket comfort and expedite the fitment process.

Methods

16 Interlink FSR-400 force-sensing resistors (FSRs) are integrated into a standard prosthetic sock by stitching them with soldered 30 AWG wire. Sensors are spaced out evenly to provide equal pressure distribution coverage. These 16 sensors connect to a custom PCB that uses an ESP32 to acquire and wirelessly transmit these measurements over Wi-Fi to a MATLAB simulation to map and display sensor readings over a 3D residual-limb model.

This system was iteratively designed, with improvements made to the MATLAB application to provide clinically-relevant metrics, the PCB for a more compact form factor, and the sensor placement for better mapping. Bench testing was performed using a 3D-printed residual limb for sensor fitment and sensor calibration was planned by correlating applied pressure to sensor response.

Results

The current prototype maps pressure distribution across 16 sensor locations and maps them onto a 3D residual-limb model.

Lab testing has confirmed that sensor response correlates with applied pressure in both location and intensity, over 50 sustained and cyclical load tests.

Clinical feedback has confirmed that the MATLAB simulation is useful and development of a custom application has begun, with the ability to record and replay clinical tests.

Conclusions

A wireless, comfortable, affordable prosthetic sensor array has been developed in the form of a sock that clinicians can use in a clinic to expedite the prosthetic fitment procedure.

Next steps include finishing sensor calibration and developing the software application.

References

[1] S. Turner and A. H. McGregor, "Perceived Effect of Socket Fit on Major Lower Limb Prosthetic Rehabilitation: A Clinician and Amputee Perspective," *Archives of Rehabilitation Research and Clinical Translation*, vol. 2, no. 3, p. 100059, 2020, doi: 10.1016/j.arrct.2020.100059.

Figures

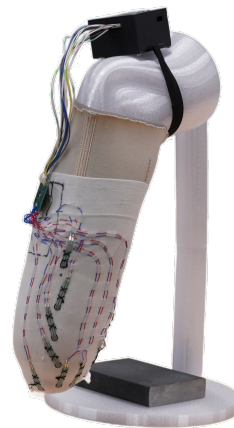


Figure 1. Prototype Sensor Array with 16 FSRs stitched onto a sock on a 3D-printed limb